Circulation

ORIGINAL RESEARCH ARTICLE

Adolescent Cardiorespiratory Fitness and the Trade-Off Between Atrial Fibrillation Risk and Cardiovascular Benefits: A Nationwide Sibling-Controlled Cohort Study

Marcel Ballin, PhD; Axel C. Carlsson, PhD*; Per Wändell, MD, PhD*; Viktor H. Ahlqvist, PhD; Mattias Brunström, MD, PhD; Pontus Henriksson, PhD; Anna Nordström, MD, PhD; Peter Nordström, MD, PhD

BACKGROUND: Young athletes and adolescents with high cardiorespiratory fitness appear to have a higher risk of atrial fibrillation (AF), but the extent to which this reflects causal effects or shared genetic, behavioral, and environmental factors remains uncertain.

METHODS: This cohort study with sibling control analysis comprised Swedish men who participated in mandatory military conscription examinations from 1972 to 1995 and completed cardiorespiratory fitness testing. The outcomes were AF and non-AF cardiovascular disease (CVD; eg, stroke and ischemic heart disease), defined as a composite end point of diagnosis or death in the National Patient Register and the Cause of Death Register, until December 31, 2023. Flexible parametric survival models estimated standardized cumulative risk differences (RDs) by deciles of fitness.

RESULTS: Among 1 124 049 men (mean age, 18.3 years), 45 179 (4.0%) had a first AF event and 96 404 (8.6%) had a first non-AF CVD event at a median age of 54.8 and 54.4 years. In population-wide analysis controlling for measured confounders, compared with the lowest decile of fitness, the highest decile had a small excess in AF that exceeded the reduction in non-AF CVD during early adulthood, whereas the reduction in non-AF CVD became larger from 45 years of age onwards. In full-sibling comparisons controlling for shared familial factors, the age-dependent trade-off disappeared entirely, leaving no age window with a net cardiovascular disadvantage. Already from 35 years of age, the reduction in non-AF CVD was larger (RD, −0.11% [95% CI, −0.21% to −0.01%]) than the excess in AF (RD, 0.06% [95% CI, −0.01% to 0.12%]). By 65 years of age, the gap further widened, with an even larger reduction in non-AF CVD (RD, −3.91% [95% CI, −5.40% to −2.42%]) compared with the excess in AF (RD, 2.30% [95% CI, 1.15% to 3.45%]).

CONCLUSIONS: High adolescent cardiorespiratory fitness is associated with a small excess in AF risk during early adulthood that is outweighed by larger reductions in non-AF CVD after controlling for familial confounders. These findings support population-level efforts to improve youth cardiorespiratory fitness and provide reassurance about the safety and benefits of high fitness levels.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: atrial fibrillation ■ cardiorespiratory fitness ■ cause of death ■ cohort studies ■ military personnel ■ myocardial ischemia ■ siblings ■ Sweden

High cardiorespiratory fitness (CRF) in youth—a reflection of physical exercise and genetic predisposition[1,2]—is recognized as a clinical vital sign and an important element in cardiovascular disease (CVD) prevention and is echoed throughout multiple guidelines and scientific statements.[3–6] At the same time, there may

Correspondence to: Marcel Ballin, PhD, Department of Public Health and Caring Sciences, Clinical Geriatrics, Uppsala University, Husargatan 3 BMC, SE 75122, Uppsala, Sweden. Email marcel.ballin@uu.se

*A.C. Carlsson and P. Wändell contributed equally.

This manuscript was handled and accepted by the editorial team led by Dr Joseph Hill.

Supplemental Material is available at https://www.ahajournals.org/doi/suppl/10.1161/CIRCULATIONAHA.125.078250.

Circulation is available at www.ahajournals.org/journal/circ

Clinical Perspective

What Is New?

- In this nationwide study of 1.1 million men followed up for up to 5 decades, high levels of adolescent cardiorespiratory fitness were associated with a small excess in the risk of atrial fibrillation during early adulthood, but this risk was outweighed by larger reductions in non–atrial fibrillation cardiovascular disease from 45 years of age onwards.
- However, when controlling for genetic, behavioral, and environmental factors shared between full siblings, the age-dependent trade-off disappeared entirely, leaving no age window with a net cardiovascular disadvantage.

What Are the Clinical Implications?

- The reduction in non–atrial fibrillation cardiovascular disease associated with high adolescent cardiorespiratory fitness exceeds the small excess in the risk of early adulthood atrial fibrillation once shared familial factors are accounted for, suggesting net cardiovascular benefits across the life course.
- These findings support population-level efforts to improve youth cardiorespiratory fitness and provide reassurance about the safety and benefits of high fitness levels.

Nonstandard Abbreviations and Acronyms

AF	atrial fibrillation
BMI	body mass index
CRF	cardiorespiratory fitness
CVD	cardiovascular disease
RD	risk difference

be potential harms from excessive exercise or high CRF such as a higher risk of atrial fibrillation (AF), which continues to fuel debate.[7,8]

According to systematic reviews, athletes, especially young athletes, have a higher risk of AF compared with nonathletes.[9,10] Some evidence also suggests an excess risk in the general population.[11,12] Two studies in Sweden found that 18-year-old men with high CRF who were followed up until their mid-40s had a higher risk of AF.[11,12] However, whether the association is causal or confounded is difficult to determine because most of the evidence comes from observational studies. Given that both CRF and AF are partly heritable,[2,13] it is possible that there may be a partial overlap in genetic variants influencing CRF levels and variants predisposing individuals to AF.[14] If so, such genetic pleiotropy could therefore create the appearance that higher CRF increases AF risk.

One approach to address this is to compare less and more fit full siblings, who share on average 50% of their genetic makeup, as well as many behavioral and environmental factors.[15] If excess AF risk diminishes within families, it may suggest that the association is driven by familial confounding rather than CRF itself. Resolving this challenge could improve our understanding of the net effects of high adolescent CRF on cardiovascular health across the life course and inform guidelines and public health policy, particularly given the rise in population-level initiatives targeting youth CRF in response to declining fitness levels.[16–18]

Here, we examined the time-dependent associations between adolescent CRF and risk of AF and non-AF CVD in 1.1 million men followed up for up to 5 decades, by linking Swedish nationwide military conscription data on CRF with health and population registers. Unlike prior studies that relied on standard analytical approaches, we also incorporated sibling comparisons in nearly half a million full siblings to control for genetic and environmental factors shared between full siblings.[19]

METHODS

Design

This cohort study was conducted by cross-linking data from nationwide registries in Sweden using the Personal Identification Number, which is unique for all Swedish residents and assigned on birth or immigration. The study was approved by the Regional Ethical Review Board in Umeå and the Swedish Ethical Review Authority (No. 2010-113-31M), which waived the need for informed consent. The study is reported according to the RECORD (Reporting of Studies Conducted Using Observational Routinely-Collected Health Data) guidelines (Supplemental File 1).[20]

Data Sources

From the Swedish Military Service Conscription Register,[21] we identified all men who participated in military conscription examinations from 1972 to 1995 and who had completed CRF testing. During this period, conscription at ≈18 years of age was mandatory for all men in Sweden. Exemptions included specific functional disabilities, routine care needs, specific religious beliefs, or conscripting elsewhere, and ≈90% of the male population conscripted during this period.[21] Using the Multi-Generation Register, we identified all conscripts who were full siblings, defined as those having the same mother and father.[22] Using the National Patient Register, we collected data on AF and non-AF CVD diagnoses from specialized outpatient and inpatient care.[23,24] Data on deaths were collected from the National Cause of Death Register.[25] Emigration and parental socioeconomic data were obtained from databases managed by Statistics Sweden. Reporting to these registers is mandatory according to Swedish law.[22,26]

Derivation of Study Population

Between 1972 and 1995, a total of 1 249 131 men conscripted. Of these, 33 645 (2.7%) were excluded because of missing CRF data, 72 042 (5.8%) because of missing covariate data, and 19 395 (1.5%) because of extreme CRF values (see details given later). Individuals with missing CRF or covariate data were excluded, consistent with a complete-case analysis approach. Thus, a total of 1 124 049 men were included in the cohort analysis (90.0% retained), of whom 477 453 were full brothers from 219 304 families and were subsequently included in the sibling analysis (Figure 1).

Assessment of CRF

CRF during conscription was assessed with a maximal ergometer bicycle test and according to a standardized protocol.[21] Briefly, conscripts performed the bicycle test after presenting with normal ECG. They began cycling for 5 minutes at a cadence of 60–70 rpm using a light external resistance determined according to body weight. The external resistance was then gradually increased by 25 W/min until voluntary exhaustion. The test results were recorded as Watt maximum (Wmax). The Wmax is the "raw measure" by which conscripts are compared during their military service, and it was used as the main exposure in this study, similar to previous studies relying on the Swedish conscription.[11,27–30] The Wmax has been found to have a strong correlation (r=0.88) with maximal oxygen uptake (ie, the gold standard metric of CRF).[31] We considered values <100 Wmax as likely due to data entry errors, and we subsequently excluded these conscripts, similar to previous studies.[27,28,32]

Ascertainment of Outcomes

The co–primary outcomes were AF and non-AF CVD (eg, stroke and ischemic heart disease) until December 31, 2023, defined as a composite end point of diagnosis or death in the National Patient Register and the Cause of Death Register.[23–25] A diagnosis of AF or non-AF CVD was ascertained as the date of first hospitalization or visit in specialized outpatient care using diagnostic codes in a primary or secondary position, and a death was classified as AF-related or non-AF CVD related if it was recorded as the primary or secondary cause on the

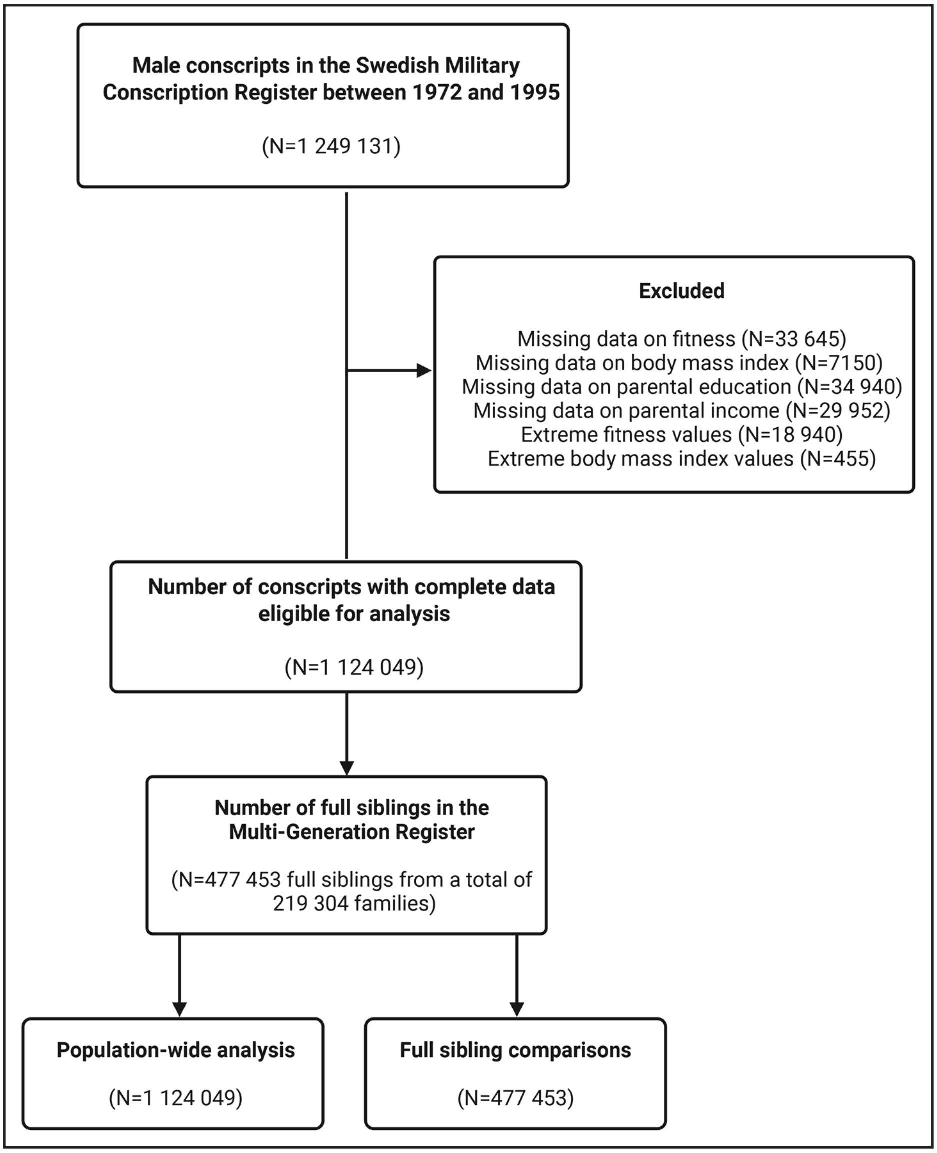

Figure 1. Participant flowchart.
Created in BioRender. V.H. Ahlqvist (2026) https://BioRender.com/eg9pufo.

death certificate. Diagnostic codes used for ascertainment of AF and non-AF CVD are given in Table S1. The positive predictive values in the National Patient Register are high for most CVDs such as stroke (69%–99%), angina pectoris (95%), heart failure (62%–88%), myocardial infarction (88%–100%), and AF (81%–97%).[23,24] Sensitivity is generally high for CVDs, including for AF (80%), myocardial infarction (77%–92%), and stroke (84%–95%), but lower for diseases typically diagnosed in primary care such as hypertension.[23,24]

Covariates

We adjusted our main analysis for measured covariates that we assumed to be causally related to both CRF and the outcomes (Figure S1). Data on age at conscription, year of conscription, and body mass index (BMI) were obtained from the Swedish Military Conscription Register. Age at conscription was included to account for between-individual differences in age at exposure assessment, which may be associated with both CRF and later outcomes, whereas year of conscription was included to control for potential cohort effects. The BMI was calculated from height and weight measured as part of the medical examination during conscription. We considered BMI values ≤15 and ≥60 kg/m^2 as extreme and likely due to data entry errors, and we subsequently excluded conscripts with such values, as in previous studies.[27,32] We collected socioeconomic data of both mothers and fathers of the conscripts from databases managed by Statistics Sweden.[26] For education, we used the highest level attained during the parent's lifetime. For income, we extracted annual disposable income for each year between 40 and 50 years of age. Within each parental birth year cohort, we then calculated income quintiles at each age and identified the highest observed quintile across this age range. This yielded a categorical variable reflecting the parent's peak "working life" income quintile. When values for both parents were available, we retained the highest value.

Statistical Analysis

Cause-specific flexible parametric survival models were used to examine the time-dependent associations between adolescent CRF and risk of AF and non-AF CVD, with age used as the underlying time scale to control for age-related variation in risk during follow-up.[33,34] Participants were followed up from conscription date until the date of AF or non-AF CVD (depending on the outcome being analyzed), emigration, death resulting from other causes, or end of follow-up (December 31, 2023). Cause-specific models have been recommended in etiological studies (ie, when estimating causal effects).[34] In contrast to Cox regression in which the baseline hazard cancels out, the flexible parametric regression directly models the baseline hazard, thereby enabling the computation of absolute risk measures that are more clinically interpretable.[35] We placed the knots at the 5th, 27.5th, 50th, 72.5th and 95th centiles of the uncensored log survival times, as per the recommendations by Harrell.[36] We modeled CRF as deciles, with the lowest decile used as referent. Hereafter, we refer to the lowest decile as least fit and the highest decile as most fit. To allow the effect of CRF to vary across time (years of age), we included an interaction term between a restricted cubic spline with 3 *df* of the follow-up time (33rd and 67th centiles of the distribution of the uncensored log survival times) and the CRF deciles.[33] For each CRF decile, we estimated the standardized cumulative incidence (1−survival) of AF and non-AF CVD across the entire follow-up and at 30, 35, 40, 45, 50, 55, 60, and 65 years of age. The standardization is a kind of parametric g formula used to estimate the average treatment effect whereby the risk of the outcome across the follow-up is predicted for every individual under different exposure configurations and then averaged over the marginal distribution of covariates. We then calculated risk differences (RDs) as differences between these standardized incidences and risk ratios as the ratio of the corresponding standardized incidences. All models were adjusted for age at conscription (continuous), year of conscription (1972, 1973–1977, 1978–1982, 1983–1987, 1988–1992, and 1993–1995), BMI (continuous and quadratic), parental education (compulsory school ≤9 years, secondary education, postsecondary education <3 years, postsecondary education ≥3 years), and parental income (5 categories).

Next, among full siblings, the aforementioned model was extended to a marginalized between-within model with robust SEs (accounting for the clustered nature of the data), thereby controlling for unobserved familial confounders.[37,38] Specifically, the between-within model includes an individual term for the exposure and each of the covariates, as well as a term for each of their family averages, thereby isolating the individual-level variation (within effect) from the family-level variation (between effect).[37,38] We have incorporated this method in several previous time-to-event analyses on CRF and health outcomes.[27,39–42]

Sensitivity and Supplementary Analyses

The following sensitivity analyses were conducted. We repeated the analyses after additional adjustment for measured handgrip and knee extension strength and systolic blood pressure, both of which are implicated in CVD,[27,43] and might be considered as confounders.[11] We applied the standard analysis in the sibling cohort (thus not accounting for shared factors) to examine whether the differences in estimates between population-wide analysis and sibling comparisons were more likely to be explained by selection bias than control for unobserved confounders.[38] To explore the role of AF severity in the association, we repeated the main analyses considering only AF-related mortality and non-AF CVD–related mortality as outcomes, thus excluding patient registry–recorded diagnoses. These analyses were conducted in the total sample but not among siblings because of limited statistical power. To further explore the role of AF severity in the association, we repeated the main analyses for subtypes of AF, including paroxysmal AF, persistent AF, and permanent AF. Last, in an exploratory effort to understand whether AF risk can be mitigated by avoiding other cardiometabolic complications, we described the frequency of type 2 diabetes diagnosis and dispensation of antidiabetic and antihypertensive medications during follow-up by deciles of CRF and stratified by AF using the National Patient Register and the Prescribed Drug Register. All analyses were performed with Stata MP version 16.1.

Data Availability Statement

The data in this study are not available to the public and will not be shared according to regulations under Swedish law. Researchers interested in obtaining the data may seek ethical approvals and inquire through Statistics Sweden. For further advice, see https://scb.se/en/services/ordering-data-and-statistics/

Table. Baseline Characteristics in the Total Population and in the Full-Sibling Cohort

	Total population (N=1 124 049)	Full-sibling cohort (n=477 453)
Birth year, median (IQR)	1966 (1960–1971)	1965 (1961–1970)
Age at conscription, mean±SD, y	18.3±0.7	18.3±0.7
BMI, kg/m²		
Mean±SD	21.8±2.9	21.7±2.8
BMI categories, n (%)		
Underweight (<18.5 kg/m²)	88 966 (7.9)	37 884 (7.9)
Normal weight (18.5–24.9 kg/m²)	913 586 (81.3)	390 811 (81.9)
Overweight (25.0–29.9 kg/m²)	102 267 (9.1)	41 248 (8.6)
Obesity (≥30.0 kg/m²)	19 230 (1.7)	7510 (1.6)
CRF, Wmax		
Mean±SD	277±52	277±52
Parental level of education, n (%)		
Compulsory school ≤9 y	319 242 (28.4)	139 246 (29.1)
Secondary education	505 510 (45.0)	210 246 (44.0)
Postsecondary education <3 y	123 757 (11.0)	50 210 (10.5)
Postsecondary education ≥3 y	175 540 (15.6)	77 751 (16.3)
Parental highest income, n (%)		
Category 1 (low income)	55 055 (4.9)	18 918 (4.0)
Category 2	108 836 (9.7)	41 820 (8.8)
Category 3	237 953 (21.2)	101 150 (21.2)
Category 4	338 444 (30.1)	145 735 (30.5)
Category 5 (high income)	383 761 (34.1)	169 830 (35.6)

BMI indicates body mass index; CRF, cardiorespiratory fitness; IQR, interquartile range; and Wmax, Watt maximum.

RESULTS

Baseline Characteristics

The total population comprised 1 124 049 men, with a mean±SD age at conscription of 18.3±0.7 years (Table). Most (81.3%) had normal BMI; about a quarter (26.6%) had parents with a high (postsecondary) level of education; and about a third (34.1%) had parents with a high (top category) level of annual income. Compared with men in the first CRF decile, men in higher CRF deciles were on average born later, had a higher BMI, and had parents with a higher level of education and income (Table S2). The sibling cohort comprised 477 453 full siblings, with similar baseline characteristics as in the total population (Table).

AF and Non–AF-CVD During Follow-Up

In the total population, 45 179 men (4.0%) experienced a first AF event at a median (interquartile range) age of 54.8 years (48.0–60.5 years), and 96 404 (8.6%) experienced a first non-AF CVD event at a median age of 54.4 years (48.3–59.9 years). In the sibling cohort, 18 638 men (3.9%) experienced a first AF event at a median (interquartile range) age of 54.5 years (48.0–60.1 years), and 40 407 (8.5%) experienced a first non-AF CVD event at a median of 54.2 years (48.4–59.5 years). Most (≥81.0%) were censored at end of follow-up (Table S3). The most common subtype of first AF event was unspecified AF, followed by paroxysmal AF, persistent AF, and permanent AF (Table S4).

CRF, AF, and Non-AF CVD

In population-wide analysis, higher levels of adolescent CRF were associated with a higher risk of AF and a lower risk of non-AF CVD (Figure 2). There was evidence of an age-dependent threshold at which the reduction in non-AF CVD exceeded the excess in AF (Figure 3 and Tables S5 and S6).

Compared with the least fit (bottom CRF decile), the most fit (top CRF decile) had an excess risk of AF that exceeded their reduction in non-AF CVD from 30 until 40 years of age, but absolute risks were small. By 30 years of age, the RD for AF was 0.10% (95% CI, 0.08% to 0.12%) and −0.001% (95% CI, −0.02% to 0.02%) for non-AF CVD, rising to 0.43% (95% CI, 0.38% to 0.47%) and −0.26% (95% CI, −0.32% to −0.21%) by 40 years of age. From 45 years of age, the excess in AF (RD, 0.70% [95% CI, 0.63% to 0.78%]) was outweighed by larger reductions in non-AF CVD (RD, −0.84% [95% CI, −0.94% to −0.75%]), which by 65 years of age widened to 3.42% (95% CI, 2.80% to 4.03%) for AF and −8.11% (95% CI, −8.80% to −7.42%) for non-AF CVD.

When comparing full siblings and thus further controlling for all behavioral, environmental, and genetic factors shared between them, the age-dependent trade-off in benefits versus harms disappeared. Despite the fact that the magnitude of reduction in non-AF CVD was reduced by more than half in sibling comparisons, there was also a reduction in the excess of AF. As a result, there was no longer a period in early adulthood and middle age during which the excess risk of AF was larger than the reduction in non-AF CVD. For example, already by 35 years of age, the reduction in non-AF CVD (RD, −0.11% [95% CI, −0.21% to −0.01%]) was larger than the excess in AF (RD, 0.06% [95% CI, −0.01% to 0.12%]). From that age and onward, the gap widened further. By 65 years of age, the RD for non-AF CVD was −3.91% (95% CI, −5.40% to −2.42%) and for AF was 2.30% (95% CI, 1.15% to 3.45%).

Sensitivity and Supplementary Analyses

The results were generally similar after further adjustment for muscular strength and blood pressure (Figure S2 and Table S7). Repeating the standard analysis in the sibling cohort (without controlling for shared factors)

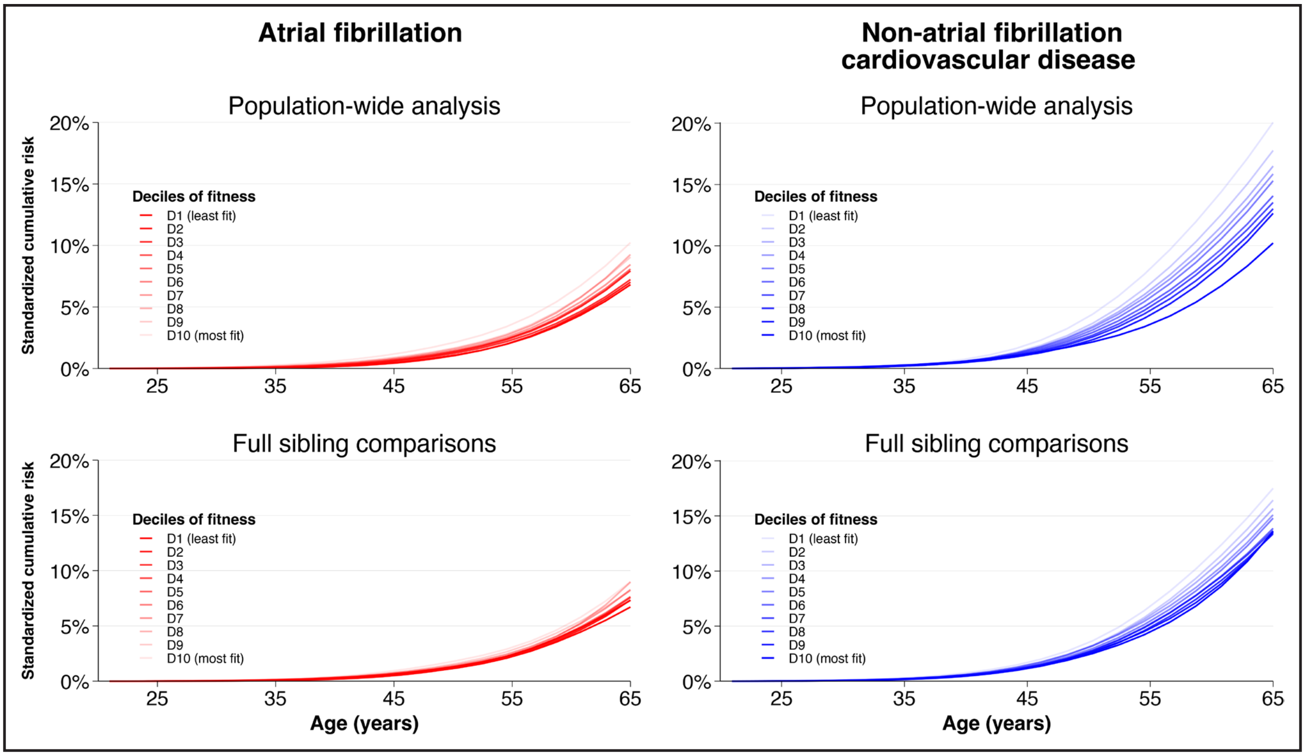

Figure 2. Standardized cumulative risk of atrial fibrillation and non–atrial fibrillation cardiovascular disease across the follow-up, by deciles of cardiorespiratory fitness in population-wide analysis and using full-sibling comparisons.

Estimates were obtained with flexible parametric survival models, extended to a marginalized between-within model in the sibling cohort, with baseline knots placed at the 5th, 27.5th, 50th, 72.5th, and 95th centiles of the uncensored log survival times and using age as the underlying time scale. All models were adjusted for age at conscription, year of conscription, body mass index, parental education, and parental income. CIs for the incidence curves were omitted for clarity because they are reflected in Table S5.

produced similar results as in the population-wide analysis in the total sample (Figure S3 and Table S8). In the analysis restricted to mortality events, there were 749 events (0.07%) of AF-related mortality and 20 895 (1.9%) events of non-AF CVD related mortality in the total population. High CRF was not associated with higher AF-related mortality (Table S9). In subtype-specific analyses, high CRF was associated with an excess in risk of paroxysmal AF but not permanent AF (Table S10). Last, dispensation of antihypertensives during follow-up was more common among individuals with a subsequent AF event compared with those without an AF event, more so among those in the highest CRF decile compared with those in the lowest decile. In contrast, no such differences were observed for diagnoses of type 2 diabetes or dispensation of antidiabetic medications (Table S11).

DISCUSSION

In this study of 1.1 million men followed up for up to 5 decades, high levels of adolescent CRF were associated with a small excess in AF risk during early adulthood that became outweighed by larger reductions in non-AF CVD by midlife and onward. However, when we controlled for genetic, behavioral, and environmental factors shared between full siblings, this age-dependent trade-off disappeared entirely, leaving no age window with a net cardiovascular disadvantage.

Unlike prior studies on CRF and AF that relied on standard analytical approaches and had a shorter follow-up,[11,12] we assessed the long-term, time-dependent associations to carefully examine when the AF excess may be outweighed by larger reductions in non-AF CVD and whether this pattern persisted after controlling for familial confounders. Doing so, our study revealed critical findings. Whereas conventional analysis suggested an age-dependent trade-off in excess risk of early AF and reduction in non-AF CVD in midlife, this pattern disappeared entirely after controlling for genetic, behavioral, and environmental factors shared between full siblings. Specifically, in these analyses, there was no longer a period during which high adolescent CRF conferred an excess in AF that exceeded the reduction in non-AF CVD, despite the fact that the magnitude of reduction in non-AF CVD became notably smaller compared with the population-wide analysis. Although the AF excess did not attenuate completely, it is worth underscoring that the absolute risks were small. Moreover, our sensitivity analyses showed an excess risk in paroxysmal AF but no excess in permanent AF or overall AF-related mortality. These findings might indicate that the observed excess in the composite AF outcome may have been driven by

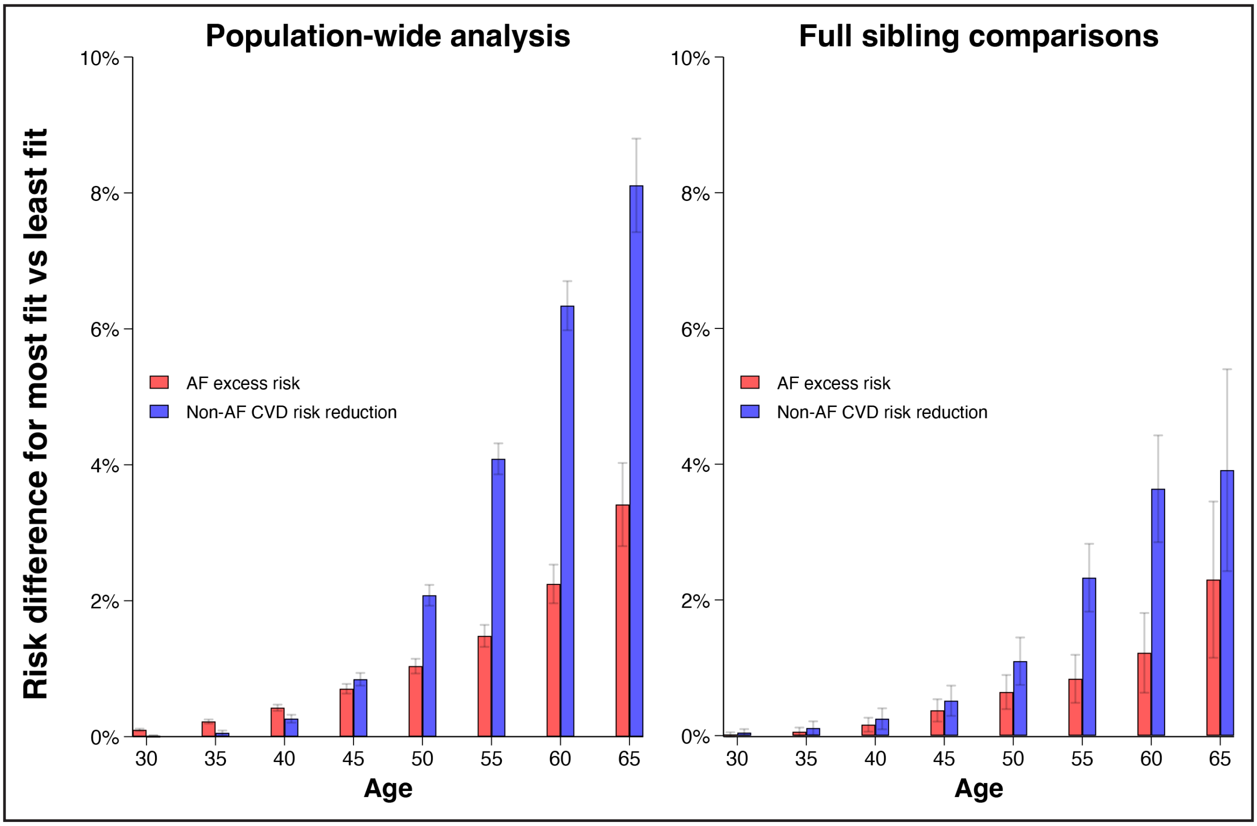

Figure 3. Differences in the standardized cumulative risk of AF and non-AF CVD from 30 to 65 years of age, comparing those with the highest (top decile) and lowest (bottom decile) levels of cardiorespiratory fitness in population-wide analysis and using full-sibling comparisons.

Estimates were obtained with flexible parametric survival models, extended to a marginalized between-within model in the sibling cohort, with baseline knots placed at the 5th, 27.5th, 50th, 72.5th, and 95th centiles of the uncensored log survival times and using age as the underlying time scale. The effect of fitness was allowed to vary across time using an interaction between a restricted cubic spline with 3 *df* the follow-up time (33rd and 67th centiles of the distribution of the uncensored log survival times) and the fitness deciles. Models were adjusted for age at conscription, year of conscription, body mass index, parental education, and parental income. Error bars show 95% CIs. AF indicates atrial fibrillation; and CVD, cardiovascular disease.

milder cases, although they should be interpreted considering that the diagnostic validity of AF subtypes in the National Patient Register is unclear.

We cannot pinpoint exactly which factors explain the discrepancy between population-wide analysis and sibling comparisons. Because of the sizeable heritable component in both CRF and CVD, including AF,[2,13,44,45] genetic pleiotropy may be a partial explanation, although this requires further investigation. Behavioral factors such as care seeking may also be involved. As shown in other work on CRF and cancer,[41] associations between CRF and generally milder health outcomes may to some extent be due to detection bias. This bias could stem from a higher degree of care seeking and health consciousness among fit people, which could be partially mitigated by controlling for shared familial factors.

Implications

The findings of this study may have implications for health promotion and cardiovascular prevention. It is important to note that after accounting for familial confounders, the cardiovascular benefits associated with high adolescent CRF outweigh any AF risk across the life course. This supports the continuation and strengthening of population-level initiatives implemented in response to declining youth CRF.[16–18] The findings may also be relevant for recommendations on exercise for AF prevention. Although current guidelines such as those from the European Society of Cardiology and the American College of Cardiology/American Heart Association emphasize the benefits of exercise and CRF in relation to AF risk, uncertainties remain concerning the role of high-intensity exercise and high levels of CRF specifically.[5,6] Our findings add reassuring evidence on the safety and benefits of high adolescent CRF, which may alleviate concerns about potential long-term arrhythmic harm.

Limitations and Strengths

Some limitations of this study should be considered. First, our findings may not be generalizable to women because only men were included. There is limited and somewhat conflicting evidence on CRF and AF risk in women,[46] although a recent study did find an excess risk among elite female endurance athletes compared with nonathletes.[47] Second, we examined the Swedish population, and similar studies in other populations are warranted. Third, we could not adjust our analysis for certain relevant factors such as alcohol consumption, smoking, and body composition (including adipose and lean tissue distribution), although they may be somewhat shared between full siblings. Similarly, we did not have data available on physical activity, which would have been valuable to examine its potential role as a confounder. Fourth, in our study, we examined the role of adolescent CRF in AF and non-AF CVD. It would be valuable for future studies to examine how variations in CRF across the life course could influence the risk of these outcomes. Fifth, this study could not determine the mechanisms underlying the observed associations, which remains an important area for future studies. Although exploratory, our descriptive analysis may suggest that such studies should aim to incorporate additional variables beyond traditional, lifestyle-related risk factors such as markers of autonomic imbalance, atrial electrophysiological alterations, and exercise-related structural remodeling such as atrial fibrosis. Sixth, although sibling comparisons enable control for all factors shared between siblings, the method rests on assumptions such as the absence of nonshared confounders and measurement error.[19] Seventh, it cannot be ruled out that some outcomes classified as non-AF CVD in our analysis were actually due to undiagnosed AF wherein AF-related sequelae eventually led to a non-AF CVD event. Whether such bias would be nondifferential or differential based on CRF levels is difficult to assess. However, as discussed previously, potential differences between more and less fit individuals in terms of care-seeking behavior may have partially been accounted for in the sibling comparison analysis.

There are also strengths, including the use of CRF data derived from standardized testing in a large population-based sample of young men, the prospective ascertainment of outcomes across several decades with virtually no attrition, and the use of sibling comparisons to reduce unmeasured familial confounding. This makes it the largest and most genetically informed study on the relationship between early-life CRF and AF.

Conclusions

Among young men with high CRF, there is a small risk of early AF that is outweighed by larger reductions in non-AF CVD once familial confounders are accounted for. These findings support population-level efforts to improve youth CRF and provide reassurance about the safety and benefits of high fitness levels.

ARTICLE INFORMATION

Received October 30, 2025; accepted April 24, 2026.

Affiliations

Department of Public Health and Caring Sciences, Clinical Geriatrics (M. Ballin, V.H.A., P.N.), and Department of Medical Sciences, Rehabilitation Medicine (A.N.), Uppsala University, Sweden. Department of Neurobiology, Care Sciences and Society, Division of Family Medicine and Primary Care (A.C.C., P.W.) and Institute of Environmental Medicine (V.H.A.), Karolinska Institutet, Stockholm, Sweden. Academic Primary Health Care Centre, Region Stockholm, Sweden (A.C.C.). Center for Primary Health Care Research, Lund University, Malmö, Sweden (P.W.). Department of Biomedicine, Aarhus University, Denmark (V.H.A.). Department of Public Health and Clinical Medicine, Umeå University, Sweden (M. Brunström). Department of Health, Medicine and Caring Sciences, Linköping University, Sweden (P.H.). School of Sport Sciences, UiT, The Arctic University of Norway, Tromsø (A.N.).

Author Contributions

M. Ballin and P.N. had full access to all the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. M. Ballin, A.C.C., P.W., and V.H.A. conceived the study. All authors contributed to the design of the study. P.N. obtained the ethical approval and data. P.N., V.H.A., and M. Brunström performed data managing. M. Ballin performed the statistical analyses. P.N. verified the underlying data. M. Ballin, A.C.C., and P.W. drafted the manuscript. All authors interpreted the results, critically revised the article for important intellectual content, read and approved the final version, and had final responsibility for the decision to submit for publication.

Sources of Funding

None.

Disclosures

Dr Ballin reports being employed at the Swedish Medical Products Agency, SE-751 03 Uppsala, Sweden. The views expressed in this article do not necessarily represent the views of this government agency. Dr Carlsson reports receiving payments for expert opinions from Link Medical. Dr Ahlqvist reports receiving lecture honoraria from Angelini Pharma. Dr Brunström reports serving on advisory boards for AstraZeneca and Amarin and receiving lecture honoraria from AstraZeneca, Amarin, and Medtronic. The other authors report no conflicts.

Supplemental Material

File S1
Tables S1–S11
Figures S1–S3

REFERENCES

1. Ortega FB, Ruiz JR, Castillo MJ, Sjostrom M. Physical fitness in childhood and adolescence: a powerful marker of health. *Int J Obes (Lond)*. 2008;32:1–11. doi: 10.1038/sj.ijo.0803774
2. Schutte NM, Nederend I, Hudziak JJ, Bartels M, de Geus EJ. Twin-sibling study and meta-analysis on the heritability of maximal oxygen consumption. *Physiol Genomics*. 2016;48:210–219. doi: 10.1152/physiolgenomics.00117.2015
3. Ross R, Blair SN, Arena R, Church TS, Despres JP, Franklin BA, Haskell WL, Kaminsky LA, Levine BD, Lavie CJ, et al; American Heart Association Physical Activity Committee of the Council on Lifestyle and Cardiometabolic Health. Importance of assessing cardiorespiratory fitness in clinical practice: a case for fitness as a clinical vital sign: a scientific statement from the American Heart Association. *Circulation*. 2016;134:e653–e699. doi: 10.1161/CIR.0000000000000461
4. Raghuveer G, Hartz J, Lubans DR, Takken T, Wiltz JL, Mietus-Snyder M, Perak AM, Baker-Smith C, Pietris N, Edwards NM; American Heart Association Young Hearts Athero, Hypertension and Obesity in the Young Committee of the Council on Lifelong Congenital Heart Disease and Heart Health in the Young. Cardiorespiratory fitness in youth: an important marker of health: a scientific statement from the American Heart Association. *Circulation*. 2020;142:e101–e118. doi: 10.1161/CIR.0000000000000866
5. Van Gelder IC, Rienstra M, Bunting KV, Casado-Arroyo R, Caso V, Crijns H, De Potter TJR, Dwight J, Guasti L, Hanke T, et al. 2024 ESC guidelines for the management of atrial fibrillation developed in collaboration with the European Association for Cardio-Thoracic Surgery (EACTS). *Eur Heart J*. 2024;45:3314–3414. doi: 10.1093/eurheartj/ehae176
6. Joglar JA, Chung MK, Armbruster AL, Benjamin EJ, Chyou JY, Cronin EM, Deswal A, Eckhardt LL, Goldberger ZD, Gopinathannair R, et al; Peer Review Committee Members. 2023 ACC/AHA/ACCP/HRS guideline for the diagnosis and management of atrial fibrillation: a report of the American College of Cardiology/American Heart Association Joint Committee

on Clinical Practice Guidelines. *Circulation*. 2024;149:e1–e156. doi: 10.1161/CIR.0000000000001193

7. Zacher J, Filipovic K, Predel G, Schmidt T. Exercise and atrial fibrillation: the dose makes the poison? A narrative review. *Int J Sports Med*. 2024;45:17–22. doi: 10.1055/a-2152-7628
8. Pham HN, Abdelnabi MH, Ibrahim R, Sainbayar E, Truong HH, Habib E, Pathangey G, Bcharah G, Singh A, Arsanjani R, et al. Exercise and atrial fibrillation: current evidence, knowledge gaps, and future directions. *Rev Cardiovasc Med*. 2025;26:39200. doi: 10.31083/RCM39200
9. Newman W, Parry-Williams G, Wiles J, Edwards J, Hulbert S, Kipourou K, Papadakis M, Sharma R, O'Driscoll J. Risk of atrial fibrillation in athletes: a systematic review and meta-analysis. *Br J Sports Med*. 2021;55:1233–1238. doi: 10.1136/bjsports-2021-103994
10. Ayinde H, Schweizer ML, Crabb V, Ayinde A, Abugroun A, Hopson J. Age modifies the risk of atrial fibrillation among athletes: a systematic literature review and meta-analysis. *Int J Cardiol Heart Vasc*. 2018;18:25–29. doi: 10.1016/j.ijcha.2018.01.002
11. Andersen K, Rasmussen F, Held C, Neovius M, Tynelius P, Sundstrom J. Exercise capacity and muscle strength and risk of vascular disease and arrhythmia in 1.1 million young Swedish men: cohort study. *BMJ*. 2015;351:h4543. doi: 10.1136/bmj.h4543
12. Crump C, Sundquist J, Winkleby MA, Sundquist K. Height, weight, and aerobic fitness level in relation to the risk of atrial fibrillation. *Am J Epidemiol*. 2018;187:417–426. doi: 10.1093/aje/kwx255
13. Roselli C, Rienstra M, Ellinor PT. Genetics of atrial fibrillation in 2020: GWAS, genome sequencing, polygenic risk, and beyond. *Circ Res*. 2020;127:21–33. doi: 10.1161/CIRCRESAHA.120.316575
14. Klevjer M, Nordeidet AN, Bye A. The genetic basis of exercise and cardiorespiratory fitness: relation to cardiovascular disease. *Curr Opin Physiol*. 2023;33:100649. doi: 10.1016/j.cophys.2023.100649
15. Ahlqvist VH, Lee BK, Chiu YH. Sibling comparisons to account for confounding in observational studies. *JAMA*. 2026;335:626–627. doi: 10.1001/jama.2025.22234
16. Brazo-Sayavera J, Silva DR, Lang JJ, Tomkinson GR, Agostinis-Sobrinho C, Andersen LB, Garcia-Hermoso A, Gaya AR, Jurak G, Lee EY, et al. Physical fitness surveillance and monitoring systems inventory for children and adolescents: a scoping review with a global perspective. *Sports Med*. 2024;54:1755–1769. doi: 10.1007/s40279-024-02038-9
17. Ortega FB, Leskosek B, Blagus R, Gil-Cosano JJ, Maestu J, Tomkinson GR, Ruiz JR, Maestu E, Starc G, Milanovic I, et al; FitBack, HELENA and IDEFICS Consortia. European fitness landscape for children and adolescents: updated reference values, fitness maps and country rankings based on nearly 8 million test results from 34 countries gathered by the FitBack network. *Br J Sports Med*. 2023;57:299–310. doi: 10.1136/bjsports-2022-106176
18. Tomkinson GR, Lang JJ, Tremblay MS. Temporal trends in the cardiorespiratory fitness of children and adolescents representing 19 high-income and upper middle-income countries between 1981 and 2014. *Br J Sports Med*. 2019;53:478–486. doi: 10.1136/bjsports-2017-097982
19. Sjölander A, Frisell T, Öberg S. Sibling comparison studies. *Annu Rev Stat Appl*. 2022;9:71–94. doi: 10.1146/annurev-statistics-040120-024521
20. Benchimol EI, Smeeth L, Guttmann A, Harron K, Moher D, Petersen I, Sorensen HT, von Elm E, Langan SM; RECORD Working Committee. The REporting of studies Conducted using Observational Routinely-collected health Data (RECORD) statement. *PLoS Med*. 2015;12:e1001885. doi: 10.1371/journal.pmed.1001885
21. Ludvigsson JF, Berglind D, Sundquist K, Sundstrom J, Tynelius P, Neovius M. The Swedish military conscription register: opportunities for its use in medical research. *Eur J Epidemiol*. 2022;37:767–777. doi: 10.1007/s10654-022-00887-0
22. Ludvigsson JF, Almqvist C, Bonamy AK, Ljung R, Michaelsson K, Neovius M, Stephansson O, Ye W. Registers of the Swedish total population and their use in medical research. *Eur J Epidemiol*. 2016;31:125–136. doi: 10.1007/s10654-016-0117-y
23. Ludvigsson JF, Andersson E, Ekbom A, Feychting M, Kim JL, Reuterwall C, Heurgren M, Olausson PO. External review and validation of the Swedish national inpatient register. *BMC Public Health*. 2011;11:450. doi: 10.1186/1471-2458-11-450
24. Everhov AH, Frisell T, Osooli M, Brooke HL, Carlsen HK, Modig K, Marild K, Lindstrom J, Skoldin K, Heurgren M, et al. Diagnostic accuracy in the Swedish national patient register: a review including diagnoses in the outpatient register. *Eur J Epidemiol*. 2025;40:359–369. doi: 10.1007/s10654-025-01221-0
25. Brooke HL, Talback M, Hornblad J, Johansson LA, Ludvigsson JF, Druid H, Feychting M, Ljung R. The Swedish cause of death register. *Eur J Epidemiol*. 2017;32:765–773. doi: 10.1007/s10654-017-0316-1
26. Ludvigsson JF, Svedberg P, Olen O, Bruze G, Neovius M. The Longitudinal Integrated Database for Health Insurance and Labour Market Studies (LISA) and its use in medical research. *Eur J Epidemiol*. 2019;34:423–437. doi: 10.1007/s10654-019-00511-8
27. Ballin M, Neovius M, Ortega FB, Henriksson P, Nordstrom A, Berglind D, Nordstrom P, Ahlqvist VH. Genetic and environmental factors and cardiovascular disease risk in adolescents. *JAMA Netw Open*. 2023;6:e2343947. doi: 10.1001/jamanetworkopen.2023.43947
28. Högström G, Nordström A, Nordström P. Aerobic fitness in late adolescence and the risk of early death: a prospective cohort study of 1.3 million Swedish men. *Int J Epidemiol*. 2016;45:1159–1168. doi: 10.1093/ije/dyv321
29. Henriksson H, Henriksson P, Tynelius P, Ekstedt M, Berglind D, Labayen I, Ruiz JR, Lavie CJ, Ortega FB. Cardiorespiratory fitness, muscular strength, and obesity in adolescence and later chronic disability due to cardiovascular disease: a cohort study of 1 million men. *Eur Heart J*. 2020;41:1503–1510. doi: 10.1093/eurheartj/ehz774
30. Herraiz-Adillo A, Ahlqvist VH, Higueras-Fresnillo S, Hedman K, Hagstrom E, Fortuin-de Smidt M, Daka B, Lenander C, Berglind D, Ostgren CJ, et al. Physical fitness in male adolescents and atherosclerosis in middle age: a population-based cohort study. *Br J Sports Med*. 2024;58:411–420. doi: 10.1136/bjsports-2023-107663
31. Andersen LB. A maximal cycle exercise protocol to predict maximal oxygen uptake. *Scand J Med Sci Sports*. 1995;5:143–146. doi: 10.1111/j.1600-0838.1995.tb00027.x
32. Ballin M, Nordstrom A, Nordstrom P. Cardiovascular disease and all-cause mortality in male twins with discordant cardiorespiratory fitness: a nationwide cohort study. *Am J Epidemiol*. 2020;189:1114–1123. doi: 10.1093/aje/kwaa060
33. Lambert PC, Royston P. Further development of flexible parametric models for survival analysis. *Stat J*. 2009;9:265–290. doi: 10.1177/1536867x0900900206
34. Austin PC, Lee DS, Fine JP. Introduction to the analysis of survival data in the presence of competing risks. *Circulation*. 2016;133:601–609. doi: 10.1161/CIRCULATIONAHA.115.017719
35. Hernan MA. The hazards of hazard ratios. *Epidemiology*. 2010;21:13–15. doi: 10.1097/EDE.0b013e3181c1ea43
36. Harrell FE. *Regression Modeling Strategies: With Applications to Linear Models, Logistic Regression, and Survival Analysis*. Springer; 2001.
37. Sjolander A. Estimation of marginal causal effects in the presence of confounding by cluster. *Biostatistics*. 2021;22:598–612. doi: 10.1093/biostatistics/kxz054
38. Sjolander A, Oberg S, Frisell T. Generalizability and effect measure modification in sibling comparison studies. *Eur J Epidemiol*. 2022;37:461–476. doi: 10.1007/s10654-022-00844-x
39. Ballin M, Nordstrom A, Nordstrom P, Ahlqvist VH. Cardiorespiratory fitness in adolescence and premature mortality: widespread bias identified using negative control outcomes and sibling comparisons [published online May 15, 2025]. *Eur J Prev Cardiol*. doi: 10.1093/eurjpc/zwaf267
40. Ballin M, Ekblom O, Nordstrom A, Ahlqvist VH, Nordstrom P. Overstated association between adolescent physical fitness and adulthood depression risk due to familial factors. *J Intern Med*. 2025;298:200–213. doi: 10.1111/joim.20109
41. Ballin M, Berglind D, Henriksson P, Neovius M, Nordstrom A, Ortega FB, Sillanpaa E, Nordstrom P, Ahlqvist VH. Adolescent cardiorespiratory fitness and risk of cancer in late adulthood: a nationwide sibling-controlled cohort study in Sweden. *PLoS Med*. 2025;22:e1004597. doi: 10.1371/journal.pmed.1004597
42. Ballin M, Ahlqvist VH, Berglind D, Brunstrom M, Herraiz-Adillo A, Henriksson P, Neovius M, Ortega FB, Nordstrom A, Nordstrom P. Cardiorespiratory fitness in adolescence and risk of type 2 diabetes in late adulthood in one million Swedish men: nationwide sibling controlled cohort study. *BMJ Med*. 2025;4:e001313. doi: 10.1136/bmjmed-2024-001313
43. Rietz H, Pennlert J, Nordstrom P, Brunstrom M. Blood pressure level in late adolescence and risk for cardiovascular events : a cohort study. *Ann Intern Med*. 2023;176:1289–1298. doi: 10.7326/M23-0112
44. McPherson R, Tybjaerg-Hansen A. Genetics of coronary artery disease. *Circ Res*. 2016;118:564–578. doi: 10.1161/CIRCRESAHA.115.306566
45. Lindgren A. Stroke genetics: a review and update. *J Stroke*. 2014;16:114–123. doi: 10.5853/jos.2014.16.3.114
46. Flannery MD, Kalman JM, Sanders P, La Gerche A. State of the art review: atrial fibrillation in athletes. *Heart Lung Circ*. 2017;26:983–989. doi: 10.1016/j.hlc.2017.05.132
47. Drca N, Larsson SC, Grannas D, Jensen-Urstad M. Elite female endurance athletes are at increased risk of atrial fibrillation compared to the general population: a matched cohort study. *Br J Sports Med*. 2023;57:1175–1179. doi: 10.1136/bjsports-2022-106035